SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 7

# Financial Analytics

Analytics in finance (sales forecasting, cash planning, credit scoring, fraud, FX risk, investment appraisal). Case 1: monthly sales forecasting with trend, season and Lebaran columns, plus nowcasting. Case 2: credit scoring with logistic regression, tree and random forest — cut-offs and fairness.

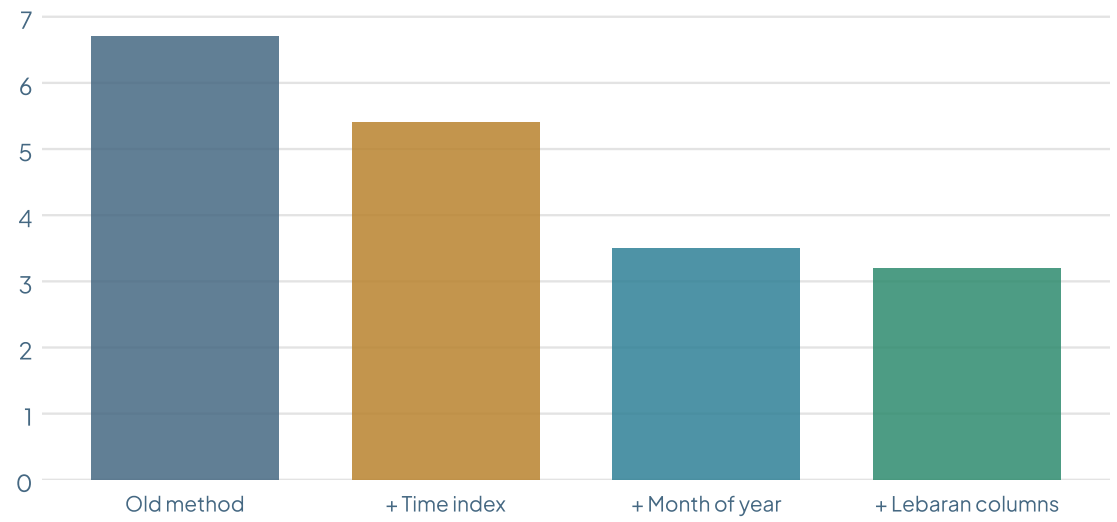
## Material

### 1 Monthly sales forecast for the 2027 budget

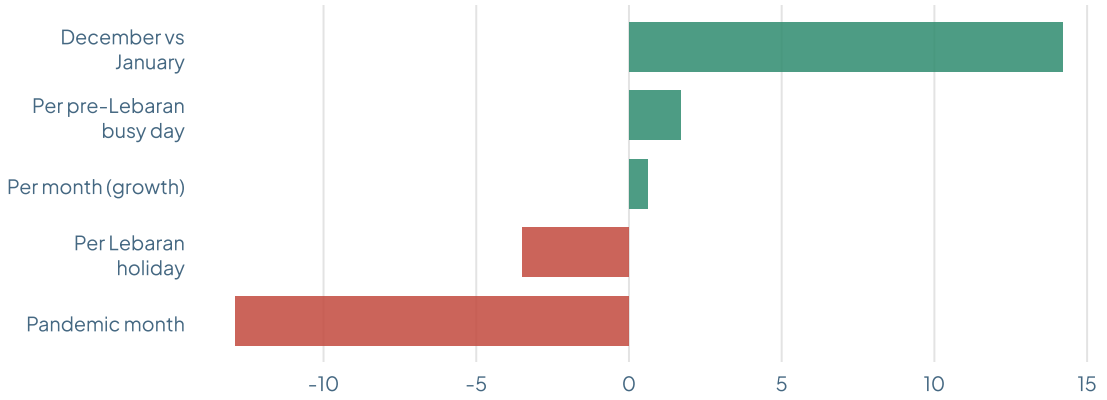
Old method error 6.7%; trend 5.4%; + month 3.5%; + Lebaran busy/holiday days 3.2%. Effects: +1.7% per busy day, −3.5% per holiday day, December +14.2%, +0.6% per month, pandemic −12.9%. For 2027 (Lebaran ~10 March) the peak moves to February — without Lebaran columns IDR 14bn lands in the wrong month.

### 2 The 2027 budget forecast in charts Deep dive

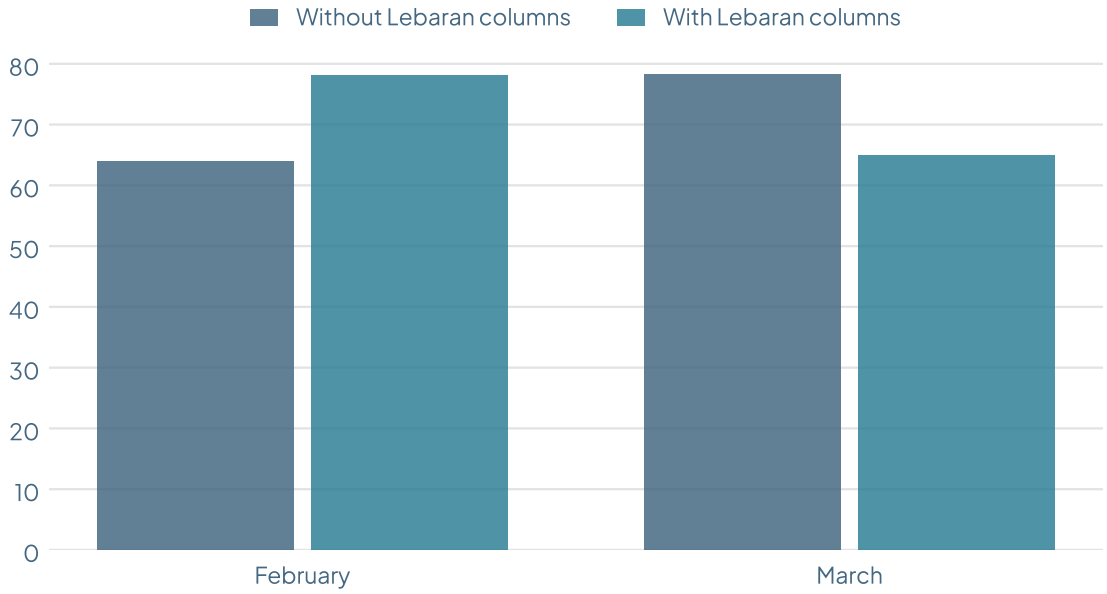
**Forecast test error by step (%)**



**Estimated effect per column (%)**




**Feb & Mar 2027 forecast (IDR bn)**

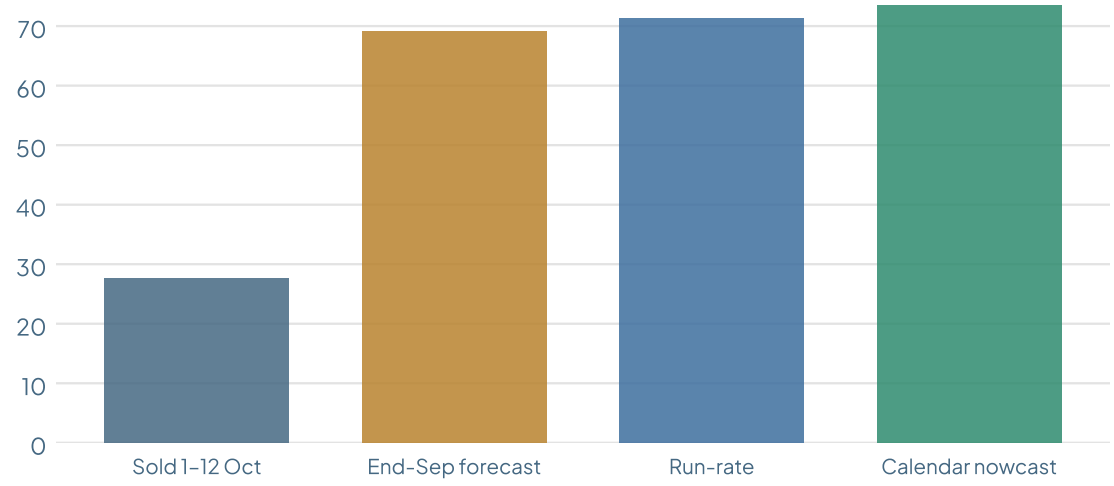



With Lebaran around 10 March 2027, the peak moves to February.

## 3 Nowcasting the current month

Run-rate 27.6 × 31/12 = 71.3; calendar-based 27.6 / 0.375 = 73.5; forecast 69.2. From day 5 the calendar nowcast beats the forecast.

## 4 Three ways to estimate October on the 12th Deep dive



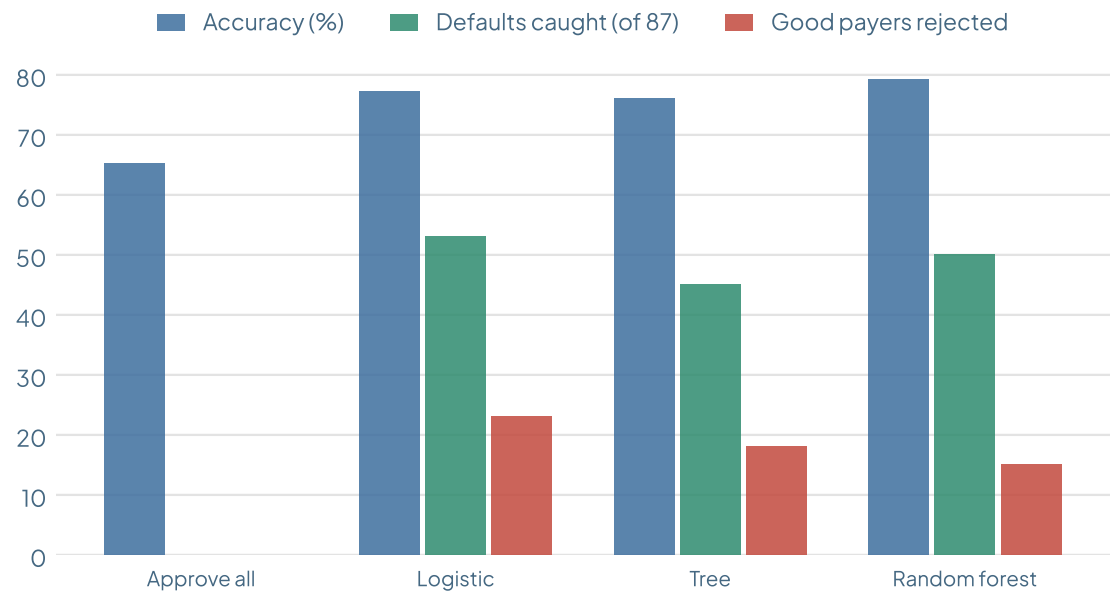

By day 12 a typical October has 37.5% of sales, not 38.7%.

## 5 Credit scoring

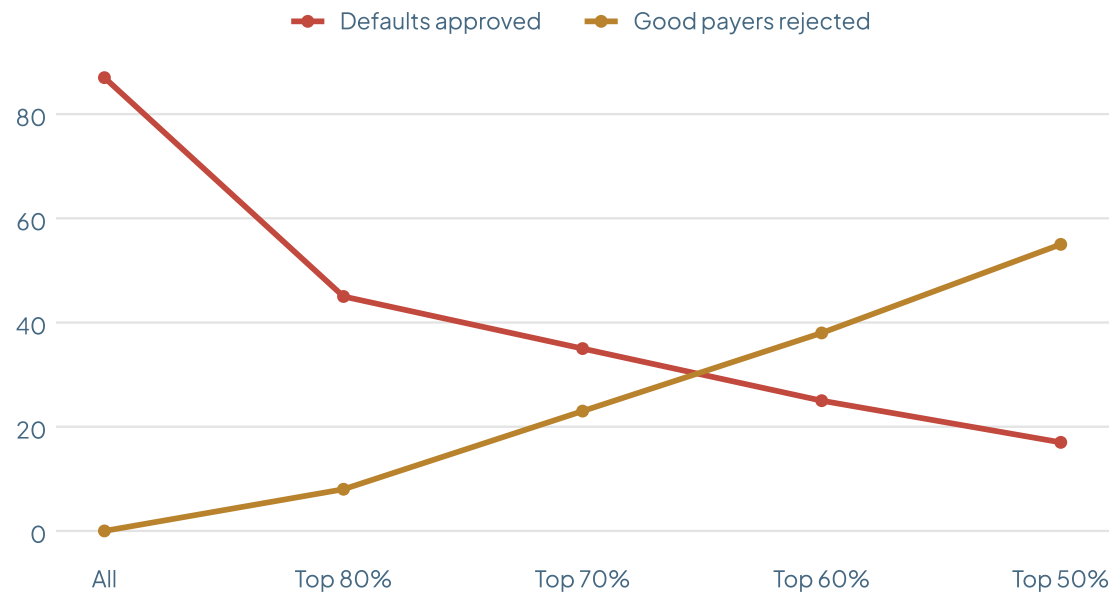
999 applications (34.9% default), 749 train / 250 test. Logistic 77.2% (53/87 defaults caught), tree 76.0%, random forest 79.2% (fewest good customers rejected). Trees read as rules. The model ranks; management sets the cut-off (top 70% approved → 20% default). Using sex adds 3.2 points of accuracy — keeping it is a policy decision.

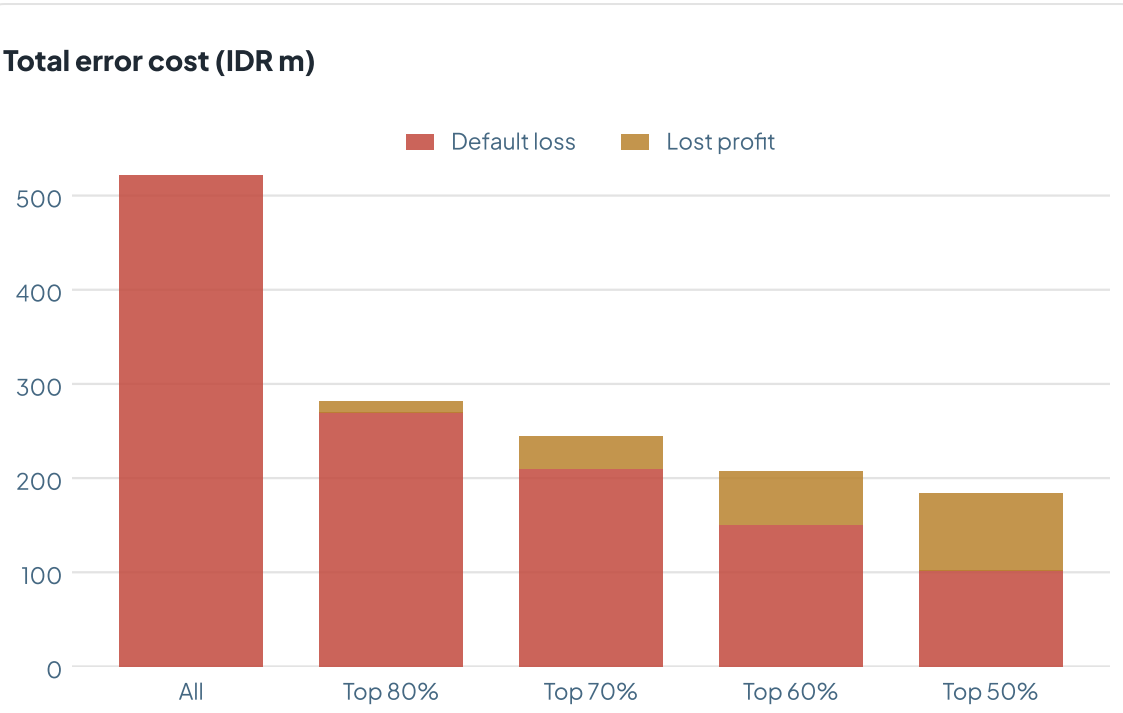
## 6 Credit scoring: methods and cut-offs in charts Deep dive

### Method comparison (250 test cases)




### Cut-off trade-off






Top 60% (207m) beats top 70% (244.5m) — and top 50% is cheaper still (184.5m).

> **!** **Fairness**
> Women 48.9% vs men 28.7% default here; using gender is an ethical and legal choice.

## 7 Measuring forecast accuracy and reading the 80% range

The "average error 3.2%" in the distributor case is the **MAPE** (mean absolute percentage error): the average absolute gap between forecast and actual, as a percentage of actual.

```
MAPE = (1/n) × Σ |actual - forecast| ÷ actual × 100%
```

| Month | Actual (IDR bn) | Forecast | \|Gap\| ÷ actual |
|---|---|---|---|
| 1 | 64 | 60 | 4 ÷ 64 = 6.25% |
| 2 | 70 | 73 | 3 ÷ 70 = 4.29% |
| 3 | 58 | 61 | 3 ÷ 58 = 5.17% |

| | | | |
|---|---|---|---|
| 4 | 66 | 66 | 0% |
| **MAPE** | | | **3.93%** |

Other measures: **MAE** (the average gap in rupiah — easy for managers: "on average we miss by IDR 2.5bn") and **RMSE** (penalises large errors more heavily). MAPE is unsuitable when some months have sales near zero, because the divisor becomes tiny.

> ! **Always test on data the model was not trained on**
> As on the slides: the model is trained up to August 2025, then forecasts the next 12 months at once. Error on training data is almost always smaller and gives false confidence.

**Reading the 80% range.** An 80% range means: if this model were used many times, about 8 in 10 months would land inside the range and 2 in 10 outside (1 above, 1 below). For the budget, use the middle figure as the target, the lower bound to plan minimum cash, and the upper bound to make sure stock and fleet are sufficient. The further ahead, the wider the range.

**Other uses in finance** (from the slide map): *cash planning* (is there enough cash every day — similar to the MK2 Week 3 cash budget), *fraud detection* (transactions unlike the usual pattern, e.g. anomaly analysis), *currency risk* (how far the exchange rate can move — Value at Risk), and *investment appraisal* (which assumption drives NPV most — sensitivity analysis and simulation).

## Key takeaways

1. Finance uses of analytics.
2. Lebaran columns cut error to 3.2%.
3. Moving holidays need their own columns.
4. Calendar nowcasts.
5. Models rank; managers set cut-offs.
6. Sensitive features are policy choices.

## Case studies

Case & worked answers

### Exercise: scoring a new application and setting the cut-off

A 25-year-old with a 24-month loan falls in an 87.3% default leaf → reject or restructure. With illustrative costs (IDR 6m per default, 1.5m profit per good payer), approving the top 60% costs 207m vs 244.5m for top 70% — cut-offs follow the error costs.

## Worked examples

Problem 1 · Easy · **Nowcasting**

Sampai tanggal 10, penjualan bulan ini Rp40 M (bulan 30 hari). Data historis: pada tanggal 10 biasanya 30% penjualan bulanan sudah tercapai. Bandingkan run-rate dan nowcast berbasis kalender.

1. Run-rate = 40 × 30 ÷ 10 = 120 (anggap semua hari sama).
2. Kalender = 40 ÷ 0,30 = 133,3.
3. Selisih Rp13 M muncul karena awal bulan biasanya lebih sepi.

**Answer:** Run-rate Rp120 M; nowcast kalender Rp133 M (lebih akurat bila pola harian stabil).

Problem 2 Medium **Memilih cut-off kredit**

Rugi per default Rp8 juta; laba per pembayar baik Rp2 juta. Top 70%: 30 default disetujui, 20 pembayar baik ditolak. Top 60%: 22 default disetujui, 35 pembayar baik ditolak. Pilih yang mana?

1. Top 70%: 30 × 8 + 20 × 2 = 240 + 40 = 280.
2. Top 60%: 22 × 8 + 35 × 2 = 176 + 70 = 246.

**Answer:** Top 60% (biaya kesalahan Rp246 juta vs Rp280 juta).

Problem 3 Easy **MAPE**

Aktual 100, 120, 90; forecast 110, 115, 99. Hitung MAPE.

1. |error| ÷ aktual: 10/100 = 10%; 5/120 = 4,17%; 9/90 = 10%.
2. MAPE = (10 + 4,17 + 10) ÷ 3 = 8,06%.

**Answer:** MAPE ± 8,1%.

Problem 4 Medium **Lebaran effects in a forecast**

The base forecast (trend + month) for February = IDR 60bn. February has 10 busy days before Lebaran (+1.7% per day) and 3 Lebaran holiday days (−3.5% per day). Effects are multiplicative. What is February's forecast?

1. Busy-day factor = $1.017^{10} \approx 1.184$.
2. Holiday factor = $0.965^{3} \approx 0.899$.
3. Forecast = $60 \times 1.184 \times 0.899 \approx$ **IDR 63.8bn**.
4. Without the Lebaran columns, this volume would land in the wrong month — which matters for cash and stock planning.

**Answer:** About IDR 63.8bn.

Problem 5 Medium **Credit-scoring metrics from test data**

Logistic regression on 250 test applications: 87 defaults, 163 good payers. The model catches 53 defaults (rejected) and wrongly rejects 23 good payers. Compute accuracy, default recall, precision, and the default rate among approved applicants.

1. TP = 53; FN = 87 − 53 = 34; FP = 23; TN = 163 − 23 = 140.

2. Accuracy = (53 + 140) ÷ 250 = **77.2%**. Recall = 53 ÷ 87 ≈ **60.9%**. Precision = 53 ÷ (53 + 23) ≈ **69.7%**.

3. Approved = FN + TN = 34 + 140 = 174. Default rate among approved = 34 ÷ 174 ≈ **19.5%** (down from 34.8% if everyone were approved).

**Answer:** Accuracy 77.2%; recall 60.9%; precision 69.7%; default among approved 19.5%.

## Practice questions

Try answering first. The answer key is below each question.

**1. Why can't fixed monthly seasonality catch Lebaran?**

A. It moves ~11 days a year
B. Always same month
C. No data
D. Pandemic

**Answer: A.** Lunar calendar.

**2. 27.6 sold by day 12; typically 37.5% done. Nowcast…**

A. 69.2
B. 71.3
C. 73.5
D. 80.0

**Answer: C.** 27.6/0.375.

**3. Which method caught the most defaults?**

A. Random forest
B. Logistic
C. Tree
D. Approve all

**Answer: B.** 53 of 87.

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**4. Who sets the approval cut-off?**

A. The model
B. Management, using error costs
C. Customers
D. Only the regulator

**Answer: B.** Model ranks; management decides.

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**5. Top 60% total cost is…**

A. 184.5
B. 207
C. 244.5
D. 282

**Answer: B.** 150 + 57.

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### Sources

- Materi P3MD WEEK 7 — Financial Analytics (MK6)